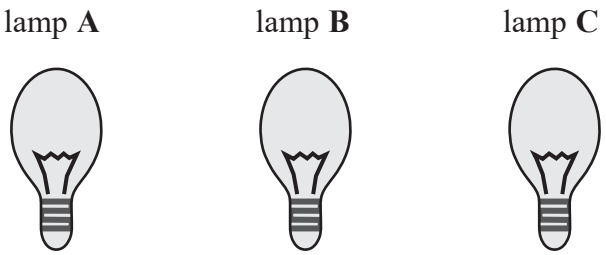
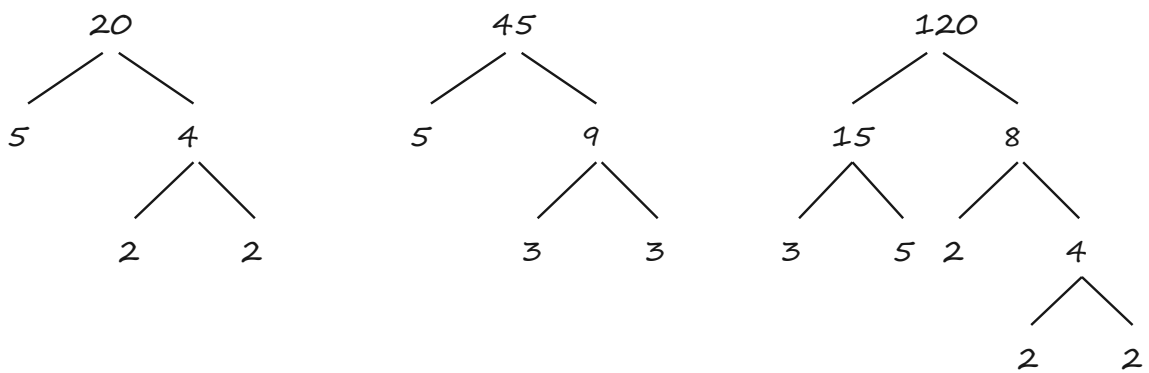
10 Here are three lamps.



Lamp **A** flashes every 20 seconds.
Lamp **B** flashes every 45 seconds.
Lamp **C** flashes every 120 seconds.

The three lamps start flashing at the same time.

How many times in one hour will the three lamps flash at the same time?



$20 = 2^2 \times 5$; $45 = 3^2 \times 5$; $120 = 2^3 \times 3 \times 5$.

1 mark
Use prime factors to find a common flashing interval

LCM = $2^3 \times 3^2 \times 5$ = 360 seconds = 6 minutes.

One hour = 3600 seconds; 3600 ÷ 360 = 10.

1 mark
Divide one hour by the common interval

1 mark
Ten further simultaneous flashes

10

(Total for Question 10 is 3 marks)